

SIU GUAN DICLOFENAC SODIUM INJECTION 25MG/ML

Diclofenac sodium 75mg/3ml Solution for injection

Description

Colourless to slightly yellow clear liquid in clear and colorless glass ampoules.

Composition

Each ml ampoule contains 25mg of Diclofenac sodium and 20mg of Benzyl alcohol as a preservative.

Action

Diclofenac sodium is a non-steroidal agent with marked analgesic/anti-inflammatory properties. It is an inhibitor of prostaglandin synthetase, (cyclo-oxygenase). Diclofenac sodium *in vitro* does not suppress proteoglycan biosynthesis in cartilage at concentrations equivalent to the concentrations reached in human beings. When used concomitantly with opioids for the management of post-operative pain, Diclofenac sodium often reduces the need for opioids.

Summary of pharmacodynamics and pharmacokinetics

Absorption

After administration of 75mg diclofenac by intramuscular injection, absorption sets in immediately, and mean peak plasma concentrations of about $2.558 \pm 0.968 \mu\text{g/ml}$ ($2.5 \mu\text{g/mL} \approx 8 \mu\text{mol/L}$) are reached after about 20 minutes. The amount absorbed is in linear proportion to the size of the dose.

Bioavailability

The area under the concentration curve (AUC) after intramuscular administration is about twice as large as it is following oral or rectal administration as this route avoids "first-pass" metabolism.

Distribution

The active substance is 99.7% protein bound, mainly to albumin (99.4%).

Diclofenac enters the synovial fluid, where maximum concentrations are measured 2-4 hours after the peak plasma values have been attained. The apparent half-life for elimination from the synovial fluid is 3-6 hours. Two hours after reaching the peak plasma values, concentrations of the active substance are already higher in the synovial fluid than they are in the plasma and remain higher for up to 12 hours.

Metabolism

Biotransformation of diclofenac takes place partly by glucuronidation of the intact molecule, but mainly by single and multiple hydroxylation and methoxylation, resulting in several phenolic metabolites, most of which are converted to glucuronide conjugates. Two phenolic metabolites are biologically active, but to a much lesser extent than diclofenac.

Elimination

Total systemic clearance of diclofenac in plasma is $263 \pm 56 \text{ mL/min}$ (mean value $\pm$ SD). The terminal half-life in plasma is 1-2 hours. Four of the metabolites, including the two active ones, also have short plasma half-lives of 1-3 hours.

About 60% of the administered dose is excreted in the urine in the form of the glucuronide conjugate of the intact molecule and as metabolites, most of which are also converted to glucuronide conjugates. Less than 1% is excreted as unchanged substance. The rest of the dose is eliminated as metabolites through the bile in the faeces.

Indications

Siu Guan Diclofenac sodium 25mg/ml for injection is indicated for treatment of rheumatoid arthritis , chronic progressive polyarthritis, acute articular rheumatism, bursitis, myositis & other inflammatory disorders.

Recommended Dosage

Deep-intramuscular injection into the gluteal area in a dose of 75mg once daily; if required in severe conditions, 75mg twice daily or prescribed by the physician.

After assessing the risk/benefit ratio in each individual patient, the lowest effective dose for the shortest possible duration should be used.

As a general recommendation, the dose should be individually adjusted. Adverse effects may be minimized by using the lowest effective dose for the shortest duration necessary to control symptoms (see section WARNINGS AND PRECAUTIONS)

Established cardiovascular disease or significant cardiovascular risk factors.

Treatment with diclofenac is generally not recommended in patients with established cardiovascular disease (congestive heart failure, established ischemic heart disease, peripheral arterial disease) or uncontrolled hypertension. If needed, patients with established cardiovascular disease, uncontrolled hypertension, or significant risk factors for cardiovascular disease (e.g. hypertension, hyperlipidaemia, diabetes mellitus and smoking) should be treated with diclofenac only after careful consideration and only at doses ≤ 100 mg daily if treated for more than 4 weeks (see section WARNINGS AND PRECAUTIONS)

As this preparation contains benzyl alcohol, its use should be avoided in children under two years of age. Not to be used in neonates.

Instruction for use

Diclofenac sodium should not be combined with other injection solution. This injection should be freshly prepared and unused portions should be discarded.

Contraindications

Diclofenac sodium is contraindicated in patients with

- hypersensitivity to Diclofenac Sodium or any of the excipients. It also contraindicated with pregnant and nursing women.
- Severe hepatic, renal and heart failure.
- Patients who have previously shown hypersensitivity reactions (e.g. asthma, angioedema, urticaria or acute rhinitis) to ibuprofen, aspirin or other non-steroidal anti-inflammatory drugs.
- Patients with active, or a history of, gastrointestinal ulcers, bleeding or perforation (two or more distinct episodes of proven ulceration or bleeding).
- History of gastrointestinal bleeding or perforation, relating to previous NSAID therapy.
- During the last trimester of pregnancy.
- Severe cardiac failure (see section WARNINGS AND PRECAUTIONS)

Warning and Precaution**Warnings**

Undesirable effects may be minimised by using the lowest effective dose for the shortest duration necessary to control symptoms .The use of Diclofenac Sodium with concomitant NSAIDs including cyclooxygenase-2 selective inhibitors should be avoided.

The elderly have increased frequency of adverse reactions to NSAIDs especially gastro intestinal bleeding and perforation which may be fatal.

Close medical surveillance is imperative in patients with symptoms indicative of gastrointestinal disorders, with a history suggestive of gastric or intestinal ulceration, with ulcerative colitis, or with Crohn's disease as these conditions may be exacerbated.

Patients with a history of GI toxicity, particularly when elderly, should report any unusual abdominal symptoms (especially Gastro intestinal (GI) bleeding).

Haematemesis, melaena, ulceration or perforation which can be fatal has been reported with all NSAIDs. They can occur at any time during treatment, with or without warning symptoms or a previous history of serious GI events. In the rare instances when gastrointestinal bleeding or ulceration occurs in patients receiving Diclofenac Sodium, the drug should be withdrawn.

The risk of GI bleeding, ulceration or perforation is higher with increasing NSAID doses, in patients with a history of ulcer, particularly if complicated with haemorrhage or perforation, and in the elderly. These patients should commence treatment on the lowest dose available. Combination therapy with protective agents (e.g. misoprostol or proton pump inhibitors) should be considered for these patients, and also for patients requiring concomitant low dose aspirin, other drugs likely to increase gastrointestinal risk.

Caution should be advised in patients receiving concomitant medications which could increase the risk of ulceration or bleeding, such as oral, corticosteroids, anticoagulants such as warfarin, selective serotonin-reuptake inhibitors (SSRIs) or anti-platelet agents such as.

Close medical surveillance is also imperative in patients suffering from impairment of hepatic function.

As with other non-steroidal anti-inflammatory drugs, allergic reactions, including anaphylactic/anaphylactoid reactions, can also occur without earlier exposure to the drug.

Like other NSAIDs, Diclofenac Sodium may mask the signs and symptoms of infection due to its pharmacodynamic properties.

RISK OF GI ULCERATION, BLEEDING AND PERFORATION WITH NSAID

Serious GI toxicity such as bleeding, ulceration and perforation can occur at any time, with or without warning symptoms, in patients treated with NSAID therapy. Although minor upper GI problems (e.g. dyspepsia) are common, usually developing early in therapy, prescribers should remain alert for ulceration and bleeding in patients treated with NSAIDs even in the absence of previous GI tract symptoms.

Studies to date have not identified any subset of patients not at risk of developing peptic ulceration and bleeding. Patients with prior history of serious GI events and other risk factors associated with peptic ulcer disease (e.g. alcoholism, smoking, and corticosteroid therapy) are at increased risk. Elderly or debilitated patients seem to tolerate ulceration or bleeding less than other individuals and account for most spontaneous reports for fatal GI events.

Precautions

Patients with renal, cardiac or hepatic impairment and the elderly should be kept under surveillance, since the use of NSAIDs may result in deterioration of renal function. The lowest effective dose should be used and renal function monitored.

The importance of prostaglandins in maintaining renal blood flow should be taken into account in patients with impaired cardiac or renal function, those being treated with diuretics or recovering from major surgery. Effects on renal function are usually reversible on withdrawal of Diclofenac Sodium.

If abnormal liver function tests persist or worsen, clinical signs or symptoms consistent with liver disease develop or if other manifestations occur (eosinophilia, rash), Diclofenac Sodium should be discontinued. Hepatitis may occur without prodromal symptoms.

Use of Diclofenac Sodium in patients with hepatic porphyria may trigger an attack.

Diclofenac Sodium may reversibly inhibit platelet aggregation. Patients with defects of haemostasis, bleeding diathesis or haematological abnormalities should be carefully monitored.

All patients who are receiving non-steroidal anti-inflammatory agents should be monitored as a precautionary measure e.g. renal function, hepatic function (elevation of liver enzymes may occur) and blood counts. This is particularly important in the elderly.

Like other drugs that inhibit prostaglandin synthetase activity, diclofenac sodium and other NSAIDs can precipitate bronchospasm if administered to patients suffering from, or with a previous history of bronchial asthma.

Appropriate monitoring and advice are required for patients with a history of hypertension and/or mild to moderate congestive heart failure as fluid retention and oedema have been reported in association with NSAID therapy.

Clinical trial and epidemiological data suggest that use of diclofenac, particularly at high dose (150mg daily) and in long-term treatment may be associated with a small increased risk of arterial thrombotic events (for example myocardial infarction or stroke).

Patients with uncontrolled hypertension, congestive heart failure, established ischaemic heart disease, peripheral arterial disease, and/or cerebrovascular disease should only be treated with diclofenac after careful consideration. Similar consideration should be made before initiating longer-term treatment of patients with risk factors for cardiovascular events (e.g. hypertension, hyperlipidaemia, diabetes mellitus, and smoking).

In patients with systemic lupus erythematosus (SLE) and mixed connective tissue disorders there may be an increased risk of aseptic meningitis.

Serious skin reactions, some of them fatal, including exfoliative dermatitis, Stevens-Johnson syndrome and toxic epidermal necrolysis, have been reported very rarely in association with the use of NSAIDs, including Diclofenac Sodium. Patients appear to be at the highest risk of these reactions early in the course of therapy, the onset of the reaction occurring in the majority of cases within the first month of treatment. Diclofenac Sodium should be discontinued at the first appearance of skin rash, mucosal lesions or any other signs of hypersensitivity.

The use of Diclofenac Sodium may impair female fertility and is not recommended in women attempting to conceive. In women who may have difficulties conceiving or who are undergoing investigation of infertility, withdrawal of Diclofenac Sodium should be considered.

Severe cutaneous reactions, including Stevens-Johnson Syndrome and toxic epidermal necrolysis (Lyell's syndrome), have been reported with diclofenac sodium. Patients treated with diclofenac sodium should be closely monitored for signs of hypersensitivity reactions. Discontinue diclofenac sodium immediately if rash occurs.

CardiovascularThrombotic Events

Observational studies have indicated that non-selective NSAIDs may be associated with an increased risk of serious cardiovascular events, principally myocardial infarction, which may increase with dose or duration of use. Patients with cardiovascular disease or cardiovascular risk of an adverse cardiovascular event in patient taking NSAID, especially in those with cardiovascular risk factors, the lowest effective dose should be used for the shortest possible duration.

There is no consistent evidence that the concurrent use of aspirin mitigates the possible increased risk of serious cardiovascular thrombotic events associated with NSAID use.

Hypertension

NSAIDs may lead to the onset of new hypertension or worsening the pre-existing hypertension and patients taking antihypertensive with NSAIDs may have an impaired anti-hypertensive response. Caution is advised when prescribing NSAIDs to patients with hypertension. Blood pressure should be monitored closely during initiation of NSAID treatment and at regular intervals thereafter.

Heart Failure

Fluid retention and oedema have been observed in some patients taking NSAIDs, there caution is advised in patients with fluid retention or heart failure.

Gastrointestinal Events

All NSAIDs can cause gastrointestinal discomfort and rarely serious, potentially fatal gastrointestinal effects such as ulcers, bleeding and perforation which may increase with dose or duration of use, but can occur at any time without warning. Caution is advised in patients with risk factors for gastrointestinal events e.g. the elderly, those with a history of serious gastrointestinal events, smoking and alcoholism. When gastrointestinal bleeding or ulcerations occur in patients receiving NSAIDs, the drug should be withdrawn immediately. Doctors should warn patient about signs and symptoms of serious gastrointestinal toxicity.

The concurrent use of aspirin and NSAIDs also increases the risk of serious gastrointestinal adverse events.

Severe Skin Reactions

NSAIDs may very rarely cause serious cutaneous adverse events such as exfoliative dermatitis, toxic epidermal necrolysis (TEN) and Stevens-Johnson Syndrome (SJS), which can be fatal and occur without warning. These serious adverse events are idiosyncratic and are independent of dose or duration of use. Patients should be advised of the signs and symptoms of serious skin reactions and to consult their doctor at the first appearance of a skin rash or any other sign of hypersensitivity.

Cardiovascular effects

Treatment with NSAIDs including diclofenac, particularly at high dose and in long term, maybe associated with an increased risk of serious cardiovascular thrombotic events (including myocardial infarction and stroke).

Treatment with diclofenac is generally not recommended in patients with established cardiovascular disease (congestive heart failure, established ischemic heart disease, peripheral arterial disease) or uncontrolled hypertension. If needed, patients with established cardiovascular disease, uncontrolled hypertension, or significant risk factors for cardiovascular disease (e.g. hypertension, hyperlipidaemia, diabetes mellitus and smoking) should be treated with diclofenac only after careful consideration and only at doses ≤ 100 mg daily when treatment continues for more than 4 weeks.

As the cardiovascular risks of diclofenac may increase with dose and duration of exposure, the lowest effective daily dose should be used for the shortest duration possible. The patient's need for symptomatic relief and response to therapy should be re-evaluated periodically, especially when treatment continues for more than 4 weeks.

Patients should remain alert for the signs and symptoms of serious arteriothrombotic events (e.g. chest pain, shortness of breath, weakness, slurring of speech), which can occur without warnings. Patients should be instructed to see a physician immediately in case of such an event.

Interactions with Other medicaments

Anti-hypertensives:

Concomitant use of NSAIDs with antihypertensive drugs (i.e. beta-blockers, angiotensin converting enzyme (ACE) inhibitors, diuretics) may cause a decrease in their antihypertensive effect via inhibition of vasodilatory prostaglandin synthesis.

Lithium and digoxin:

Diclofenac sodium may increase plasma concentrations of lithium and digoxin.

Diuretics:

Like other NSAIDs, Diclofenac sodium may inhibit the activity of diuretics. Concomitant treatment with potassium-sparing diuretics may be associated with increased serum potassium levels, which should therefore be monitored frequently.

Anticoagulants:

Although clinical investigations do not appear to indicate that Diclofenac sodium has an influence on the effect of anticoagulants, there are isolated reports of an increased risk of haemorrhage with the combined use of diclofenac and anticoagulant therapy. Therefore, to be certain that no change in anticoagulant dosage is required, close monitoring of such patients is required. As with other non-steroidal anti-inflammatory agents, diclofenac in a high dose can reversibly inhibit platelet aggregation.

Antidiabetic agents:

Clinical studies have shown that Diclofenac sodium can be given together with oral antidiabetic agents without influencing their clinical effect. However there have been isolated reports of hypoglycaemic and hyperglycaemic effects which have required adjustment to the dosage of hypoglycaemic agents.

Methotrexate:

Cases of serious toxicity have been reported when methotrexate and NSAIDs are given within 24 hours of each other. This interaction is mediated through accumulation of methotrexate resulting from impairment of renal excretion in the presence of the NSAID.

Quinolone antimicrobials:

Convulsions may occur due to an interaction between quinolones and NSAIDs. This may occur in patients with or without a previous history of epilepsy or convulsions. Therefore, caution should be exercised when considering the use of a quinolone in patients who are already receiving an NSAID.

Other NSAIDs including cyclooxygenase-2 selective inhibitors and corticosteroids:

Co-administration of Diclofenac sodium with aspirin or corticosteroids may increase the risk of gastrointestinal bleeding or ulceration. Avoid concomitant use of two or more NSAIDs.

Anti-platelet agents and selective serotonin reuptake inhibitors (SSRIs):

Increased risk of gastrointestinal bleeding.

Mifepristone:

NSAIDs should not be used for 8-12 days after mifepristone administration as NSAIDs can reduce the effect of mifepristone.

Ciclosporin and Tacrolimus:

Cases of nephrotoxicity have been reported in patients receiving concomitant ciclosporin and NSAIDs, including diclofenac sodium. Possible increased risk of nephrotoxicity when NSAIDs are given with tacrolimus. This might be mediated through renal antiprostaglandin effects of both NSAID and calcineurin inhibitor.

Cardiac glycosides:

Concomitant use of cardiac glycosides and NSAIDs in patients may exacerbate cardiac failure, reduce GFR and increase plasma glycoside levels.

Pregnancy and lactation

Diclofenac sodium should not be used during the first two trimesters of pregnancy or labour unless the potential benefit outweighs the potential risk to foetus. The lowest effective dose should be used and duration kept as short as possible.

Diclofenac sodium is distributed into breast milk that may be harmful to breast-fed infants. It should if possible be avoided when breast feeding.

Adverse Drug Reaction/ Side effects

Side effects of Diclofenac sodium are pain and occasionally tissue damage at the site of injection. Local irritation and necrosis may be experienced. Rhinorrhoea, tightness of chest, wheezing dyspnoea and anaphylactic shock on some aspirin-sensitive patients. Fatal hyponatraemia on the elderly. Aplastic anemia, haemolytic anemia, thrombocytopenia and agranulocytosis.

Others Advers effects is Dermatological : Occasional - rashes or skin eruptions Cases of hair loss, bullous eruptions, erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis (Lyell's syndrome), and photosensitivity reactions have been reported.

If serious side-effects occur, Diclofenac Sodium should be withdrawn.

Frequency estimate: frequent:> 10 %, occasional:> 1 - 10 %, rare:> 0.001 - 1 %, isolated cases: < 0.001 %.)

Gastro-intestinal tract

Occasional: Epigastric pain, other gastrointestinal disorders (e.g. nausea, vomiting, diarrhoea, abdominal cramps, dyspepsia, flatulence, anorexia).

Rare: Gastritis, gastrointestinal bleeding (haematemesis, melaena, and bloody diarrhoea), gastrointestinal ulcers with or without bleeding or perforation (sometimes fatal, particularly in the elderly) may occur.

In isolated cases: Aphthous stomatitis, glossitis, oesophageal lesions, lower gut disorders (e.g. non-specific haemorrhagic colitis and exacerbations of ulcerative colitis or Crohn's proctocolitis, colonic damage and stricture formation), pancreatitis, constipation.

Central Nervous System disorders

Occasional: Headache, dizziness, or vertigo.

Rare: Drowsiness, tiredness, hypotension.

In isolated cases: Disturbances of sensation, paraesthesia, memory disturbance, disorientation, insomnia, irritability, convulsions, depression, confusion, hallucinations, malaise, anxiety, nightmares, tremor, psychotic reactions, aseptic meningitis (especially in patients with existing auto-immune disorders, such as systemic lupus erythematosus and mixed tissue disease), with symptoms such as fever, stiff neck, headache, nausea and vomiting.

Special senses

Isolated cases: Disturbances of vision (blurred vision, optic neuritis, diplopia), impaired hearing, tinnitus, taste disturbances.

Skin

Occasional: Rashes or skin eruptions.

Rare: Urticaria.

In isolated cases: Bullous eruptions, eczema, erythema multiforme, Stevens-Johnson syndrome, Lyell's syndrome, (acute toxic epidermolysis), erythroderma (exfoliative dermatitis), loss of hair, photosensitivity reactions, purpura including allergic purpura.

Kidney

Rare: Oedema.

In isolated cases: Acute renal insufficiency, urinary abnormalities (e.g. haematuria, proteinuria), interstitial nephritis, nephrotic syndrome, papillary necrosis.

Liver

Occasional: Elevation of serum aminotransferase enzymes (ALT, AST).

Rare: Liver function disorders including hepatitis (in isolated cases fulminant) with or without jaundice.

Blood

In isolated cases: Thrombocytopenia, leucopenia, agranulocytosis, haemolytic anaemia, aplastic anaemia.

Vascular

Isolated cases: Vasculitis.

Respiratory

Isolated cases: Pneumonitis.

Cardiovascular system

Isolated cases: Palpitations, chest pain, hypertension, congestive heart failure.

Other organ systems

Isolated cases: Impotence.

Reactions to the intramuscular injection

Occasional: Reactions such as local pain and induration.

Isolated cases: Abscesses and local necrosis at the intramuscular injection site.

Hypersensitivity

Hypersensitivity reactions have been reported following treatment with NSAIDs. These consist of non-specific allergic reactions and anaphylaxis, respiratory tract reactivity comprising of asthma, aggravated asthma, bronchospasm or dyspnoea, or assorted skin disorders, including rashes of various types, pruritus, urticaria, purpura angiodema and, more rarely exfoliative and bullous dermatoses (including epidermal necrolysis and erythema multiforme).

Clinical trial and epidemiological data suggest that use of diclofenac, particularly at high doses (150 mg daily) and in long-term treatment may be associated with a small increased risk of arterial thrombotic events (for example myocardial infarction or stroke).

Cardiac disorders

Uncommon:* Myocardial infarction, cardiac failure, palpitations, chest pain.

*The frequency reflects data from long-term treatment with a high dose (150mg/day)

Description of selected adverse drug reactions

Arteriothrombotic events

Meta-analysis and pharmacoepidemiological data point toward an increased risk of arteriothrombotic events (for example myocardial infarction) associated with the use of diclofenac, particularly at a high dose (150 mg daily) and during long-term treatment (see section WARNINGS AND PRECAUTIONS)

Symptoms and treatment of Overdose

Symptoms include headache, nausea, vomiting, epigastric pain, gastrointestinal bleeding, rarely diarrhoea, disorientation, excitation, coma, drowsiness, tinnitus, fainting, occasionally convulsions. In rare cases of significant poisoning acute renal failure and liver damage are possible.

Therapeutic measures

Patients should be treated symptomatically as required. Within one hour of ingestion of a potentially toxic amount, activated charcoal should be considered. Alternatively, in adults gastric lavage should be considered within one hour of ingestion of potentially toxic amounts. Frequent or prolonged convulsions should be treated with intravenous diazepam. Other measures may be indicated by the patients clinical condition.

Presentation

This injection is available in 3ml ampoules
10 ampoules are packed into one paper box.

Storage conditions

Protect from light and store below 30°C. Keep out of reach of children.

Shelf-life

The injections can be used within 60 months from the date of manufacture if kept as recommended.
Discard any unused portion.

Registration number

SIU GUAN DICLOFENAC SODIUM INJECTION 25MG/ML - MAL08091431AZ

Manufactured by

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